Factors and Data Frames

1. Convert the following character vector to a factor and make a barplot to visualize its distribution (provide your code and sketch of the barplot).

```
opinion <- c("dislike", "neutral", "dislike", "love", "like", "like")
```

2. Add the possibility for an additional level named `"despise"`. Then, reorder the factor sensibly from most negative to most positive and remake your barplot.

The following four question deal with the `iris` data set, which is built in to R. You can view it by just typing `iris`.

3. What are the dimensions of the data frame?

4. What is the unit of observation (i.e. what is in each row)? See `?iris`.

5. What is the type of each of the atomic vectors?

6. How can you create a new data frame that has the same number of rows but excludes the columns dealing with sepals?

7. Shift the `name` column in `star_wars` to be the row names of that data frame. That's to say, use it to populate the row names and then remove it as a column.

```
star_wars <- data.frame(
    name = c("Anakin", "Padme", "Luke", "Leia"),
    gender = c("male", "female", "male", "female"),
    height = c(1.88, 1.65, 1.72, 1.50),
    weight = c(84, 45, 77, 49)
)
```